Gas Mask

Category: **Safety & PPE** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Gas Mask (full vision facemask) & Canister for use with ALP

What it is used for

A full-face gas mask with a compatible canister helps protect the wearer from specified airborne contaminants when used under an approved respiratory-protection programme. Correct selection, fit testing, inspection, and use are essential because a canister cannot protect against every gas, vapour, or oxygen-deficient atmosphere.

Main parts

- Full-face mask seals around the face and protects the eyes and respiratory system.
- Visor provides a wide field of vision.
- Inhalation valve directs incoming air through the approved airflow path.
- Exhalation valve allows exhaled air to leave the mask.
- Compatible canister or cartridge filters or treats specified contaminants.
- Head harness holds the mask in the correct position.
- Face seal forms the protective seal against the skin.
- Communication or voice diaphragm supports speech where fitted.

Before you start

1. Use the mask only under an approved respiratory-protection programme and after suitable training.
2. Identify the contaminant, concentration, oxygen level, and required protection before selecting a mask and canister.
3. Never use an air-purifying canister in an oxygen-deficient or immediately dangerous atmosphere.
4. Confirm the canister is approved for the specific contaminant and is within its service life.
5. Inspect the face seal, visor, valves, harness, connectors, and canister for damage or contamination.
6. Complete the required fit test and user seal check; facial hair must not interfere with the sealing surface.
7. Confirm the wearer can safely use the mask and that an emergency plan is available.

How to use

1. Wash and dry hands before handling the mask and canister.
2. Inspect the mask and fit the specified canister according to the manufacturer's instructions.
3. Check that valves, connectors, and seals are correctly installed and unobstructed.
4. Put on the mask and adjust the head harness evenly without overtightening.
5. Perform the prescribed user seal check before entering the designated work area.

6. Enter the area only after the responsible safety person confirms the atmosphere and protection are suitable.
7. Keep the mask correctly positioned and monitor for leakage, breathing resistance, visor fogging, or discomfort.
8. Leave the area immediately if leakage, unusual odour, dizziness, breathing difficulty, or another warning sign occurs.
9. Move to a safe area before removing the mask and follow the site decontamination procedure where required.
10. Remove the canister and clean, inspect, and store the mask according to the approved procedure.
11. Record inspections, canister changes, defects, and any use-related concerns.

⚠ Safety

- Never rely on a gas mask for an unknown atmosphere, oxygen deficiency, or conditions beyond its approved limits.
- Use only a canister approved for the identified contaminant and exposure conditions.
- Do not use damaged, expired, saturated, or incorrectly fitted canisters.
- A user seal check does not replace formal fit testing.
- Do not remove the mask in a contaminated area unless an authorised emergency procedure specifically directs it.
- If breathing becomes difficult or contamination is suspected, leave immediately and follow the emergency plan.

Care & storage

- Clean and disinfect the mask using manufacturer-approved products and methods.
- Inspect valves, face seal, visor, harness, and connectors after each use.
- Store the mask in a clean, dry, sealed container away from chemicals, sunlight, heat, and deformation.
- Replace canisters according to the approved change schedule and service-life limits.
- Remove damaged equipment from service and arrange authorised inspection or replacement.

Common mistakes to avoid

- Using a canister that is unsuitable for the contaminant or exposure level.
- Entering a hazardous atmosphere without confirming oxygen levels and protection requirements.
- Skipping fit testing or the required user seal check.
- Continuing to use a damaged mask or an exhausted canister.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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